Case study brief

Lender data pipeline

From	Head of Data and Analytics
To	Data engineering
Subject	Lender data platform: load, clean and model six source files
Scenario	Fictional South African consumer lender. All data is simulated (Nov 2024 to Oct 2026).
Due	Two weeks from kickoff

Background

Our analysts receive customer, application, loan, repayment, partner and campaign files from different systems and clean them by hand each time. Credit risk and partnership analytics both wait on this. We want one repeatable pipeline on Databricks that produces trusted tables.

The request

Load the six files unchanged, clean them, check them, and publish tables the analysts can use directly, including a one-row-per-loan table with an early-default label and a table linking each application to the campaign running for its partner.

What we need back

- Bronze, silver and gold tables in Unity Catalog, with source file and load time on every raw row.
- Cleaning rules in tested Python code, not buried in notebooks.
- Data quality checks and a short report of raw versus clean row counts and the problems found.
- A single notebook that rebuilds everything and gives the same counts on a re-run.
- A README that a new team member can follow.

Data provided

File	One row per
customers.xlsx (2 sheets)	customer
applications.csv	application
loans.csv	approved loan
repayments.csv	instalment due
partners.csv	referral partner
campaigns.csv	marketing campaign

Expect mixed date and ID formats, duplicated rows, currency stored as text, placeholder values, spelling variants and hand-entered errors.

Constraints

- Keep raw data unchanged in bronze; never fix problems there.
- Minimise personal information in silver (POPIA).

- No deleting rows silently: report what was removed and why.
- Use Databricks, Python, SQL and PySpark only.

Success looks like

- Running the pipeline twice gives identical counts.
- Every key reconciles across tables.
- Tests for the cleaning rules pass.